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Proponent Name:	Dr. Remar D. Apolinario
Academic Rank:	Asso. Professor V
Co-Authors (with academic rank):	Amel L. Magallanes, PhD, Professor VI
Campus/Satellite College/Address:	
Telephone/Cell Phone/Fax Number:	0919 981 6543
E-mail Address:	remarapolinario28@gmail.com

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(9A) Abstract

Nowadays grass cutter machines are becoming very popular, which can be seen in every daily life. The model of grass cutter regularly used nowadays is petrol grass cutter type. In this model the gasoline is converted into air gas and the powerful two-stroke engines are ignited to operate the equipment resulting to exhaustion of burnt fuel which contributes to air pollution thereby posing danger to the environment and to the operator as well. Thus, the innovation of solar power grass cutter was developed.

Specifically, this study aimed to: The study aimed to: (1) determine the charging rate of the Backpack Solar Powered Grass Cutter in terms of milliamp hours (mAH); (2) determine the capacity of discharging rate in terms of wattage used and the milliamp hour (mAH) of battery of the solar power grass cutter without solar and with solar charging; (3) determine the level of acceptability of the Solar Power Grass cutter as to its design, composition, and operating performance.

The fabrication and construction of the equipment was done in the College of Engineering, Architecture, and Technology and was verified and measured in terms of its charging rate, discharging rate at different time intervals using three trials. The design, composition, and operating performance of the system were evaluated using the adopted evaluation sheet by thirty (30) evaluators composed of 10 industrial technology academe from electrical and electronics technology, 3 registered electrical engineers, 7 casual workers from DPWH, 3 local farmers and 7 utility aide.

The charging rate of the equipment was 700 mA at 29°C - 30°C average heat temperature and it takes 10 hours to fully charge the battery when not in use. While the discharging rate is 1,525 mAH and it takes 4.59 hours of continuous used to fully drain without the support of the solar panel. On the other hand the discharging rate of battery when the solar panel is connected is only 560 mAH and it takes 12.5 hours to fully drain the battery. From the initial mAH of battery, it increases when connected to solar panel support for cutting the grass field area of 50 sq. meters. Furthermore, the equipment was "Very Acceptable" in terms of design, composition, and operating performance.

(9B) Keywords: *Grass Cutter, solar powered cutter*

(10) Introduction

Thousands of people around the globe always look for a way on how to make their work faster and easier, but sometimes they almost forgot the negative effect that it can bring to the environment and even into human health just to satisfy their needs. The researchers of the modern society, want to contribute an innovative product that came from unused parts of an automobile that sometimes consider as junks in our surroundings. As a useful citizen in society they want to innovate or create some devices that can help us in our needs without harming our environment or human health.

According to Energy Management Based on Frequency Approach for Hybrid Electric Vehicle Applications (2012), from ancient time, the sun has been the major source of energy in life on earth. The solar power was being used directly in drying clothes, curing agricultural product, preserving food, etc. Even today, the energy from fuel-wood, petroleum, paraffin, hydroelectricity, and even their food originate indirectly from the sun. Solar energy is almost unbounded. The total energy we obtain from the sun far exceeds our energy demands. Ever since the industrial revolutions, human have been dependent on fuels, electricity and wind energy. For human enlargement in many countries, there is study and trials are going on the solar power, (Energy Management Based on Frequency Approach for Hybrid Electric Vehicle Applications 2012).

This developmental research motivated the researcher to make a new concept by backpack solar-powered grass cutter from an auxiliary fan. In this concept, Users cut grasses that strew on their surroundings and in terms of agricultural industries it can help the local farmers to cut their own small plantations in lawns and gardens.

Nowadays grass cutter machines are becoming very popular. The current model of grass cutter uses

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motor fuel engine and because it utilizes fuel, it emits smoke that contributes to the rising pollution level. Likewise it is more costly and hazardous. Maintenance of such a conventional machine is much more expensive because of its motor oil and other cleaning equipment to maintain its performance. The exhausted burnt fuel from the motor engine is much more dangerous and hazardous to the human being and the environment. To avoid these shortcomings the researcher plan to build a new type of grass trimmer which is run by solar power as a source of energy and this model is also economical. The aim of the project is to make the grass trimmer that utilizes the solar power in order to save the non-renewable fuel.

The design of the project will help both the customer, end user and the environment. The development of Backpack solar powered grass cutter will help the user from cutting their own lawns in a very cheap constructing and will reduce both environmental and noise pollution as green technology.

Legal Basis for the Utilization of Renewable Energy Resources in the Philippines

In the declaration of policies of the Republic Act 9513 of 2008 also known as “An Act Promoting The Development, Utilization And Commercialization Of Renewable Energy Resources And For Other Purposes”, the utilization and commercialization of energy resources was described as (a) Accelerate the exploration and development of renewable energy resources such as, but not limited to, biomass, solar, wind, hydro, geothermal and ocean energy sources, including hybrid systems, to achieve energy self-reliance, through the adoption of sustainable energy development strategies to reduce the country's dependence on fossil fuels and thereby minimize the country's exposure to price fluctuations in the international markets, the effects of which spiral down to almost all sectors of the economy;(b) Increase the utilization of renewable energy by institutionalizing the development of national and local capabilities in the use of renewable energy systems, and promoting its efficient and cost-effective commercial application by providing fiscal and non-fiscal incentives; and (c) Encourage the development and utilization of renewable energy resources as tools to effectively prevent or reduce harmful emissions and thereby balance the goals of economic growth and development with the protection of health and the environment (Republic Act 9513, 2008).

In regards to the policy of the Republic Act of the Philippines, the researcher would like to be a useful citizen by developing an eco-friendly device utilizes renewable energy.

The present study was also developed based on the principle of alternative energy. It encompasses all those things that do not consume fossil fuel. They are widely available and environment-friendly. They cause little or almost no pollution. There have been several alternative energy projects running in various countries to reduce the dependence on traditional fossil fuels (“Alternative Energy”, 2017).

This energy source is an alternative to fossil fuel. These alternatives are intended to address concerns about fossil fuels, such as its high carbon dioxide emissions, which is an important factor in global warming. Marine energy, hydroelectric, wind, geothermal and solar power are all alternative sources of energy. The nature of what constitutes an alternative energy source has changed considerably over time because of the variety of energy choices and differing goals of their advocates (Zehner, 2012).

Related Inventions and Studies

Lawnmowers are the most common kind of grass cutter that most homeowners use. Mowers may be rotary types, where blades swirl horizontal to the ground, or reel mowers where blades cut vertically. Typically gas or electric powered, rotary mowers may be self-propelled or rely on manpower to make them move. For large areas, reel mowers are pulled behind lawn tractors. Smaller versions are manually pushed across lawns. Lawnmowers are designed to keep the grass cut short, but a good rule is to never cut off more than one-third of the height of your grass. Most mowers have levers that allow you to raise or lower the mowing blades to cut the grass to your desired length. (“uses of Grass Cutter”, n.d.)

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Types of Portable Grass Cutter

Line trimmers and **brushcutters** (sometimes collectively known as “stick trimmers”) are vital parts of the horticultural armory, but to the newcomer the sheer variety of machines on the market today can be baffling. There is, for example, a big difference between a lightweight domestic grass trimmer designed for tidying up long grass and weeds around the garden and a heavy-duty industrial brushcutter, which is better suited for quick clearance of dense scrub and undergrowth on undeveloped land.

Electric Corded Grass Trimmers the power source used by your grass trimmer or brushcutter will go a long way to determining the kind of vegetation you can cut and the size of the area you can maintain effectively. The power source will also make a big difference to your user-experience; electric machines, for instance, are quieter, cleaner and more straightforward to use than petrol machines but typically lack the power for tougher professional work.

Cordless Trimmers & Brushcutters modern cordless trimmers feature hi-tech lithium-ion batteries to help ensure effective cutting even in tricky situations that they can provide the cutting power and freedom to roam required for working in larger gardens but without the extra hassle and expense of running a petrol machine.

Petrol Grass Trimmers & Brushcutters This is because petrol grass trimmers and brushcutters are better suited to long days of constant use, making them ideal for trimming undergrowth and mowing in areas where the grass is too long or rough to maintain with a lawnmower. Typically fitted with powerful two-stroke and four-stroke engines, ranging from around 25cc to more than 50cc in commercial-grade machines, petrol trimmers, and brushcutters are what you will need when taking on serious cutting work. However, petrol trimmers range significantly in terms of size, power, and price, so the best machine for you will depend on your needs; spot trimming in large gardens and other expansive areas.

Cutting Equipment

Nylon Line nylon line trimmer heads are designed for cutting long grass and other light vegetation, so dedicated grass trimmers will come with these as standard. These days most nylon line trimmer heads also incorporate a variety of user-friendly features to enable quick loading and advancement of the cutting line.

Metal Blades feature different blade attachments that allow you to cut a variety of materials. They're generally durable and reliable machines, as long as they receive proper maintenance. However, blade life depends on what you're cutting and how often. For example, if you're working frequently in rocky areas or cutting through hard stems or branches, you may have to change or sharpen the blade daily. Before each use, you should check the blade sharpness, because working with a dull blade shortens service life and can pose a safety hazard. (“Grass Trimmer”, n.d.)

Prior Art

“The Cordless Grass Trimmer Having Removable Battery Pack of (Bartasevick, 1976) was one of the bases of the present development of the “Solar Power Grass Trimmer”.

To complete the design of the electrical system and to improve its performance the equipment was adjusted to the latest model which is much more compatible in any terrain surface and any angle you can trim the grass infrequently (Figure 1). The Cordless Grass Trimmer Having Removable Battery Pack of (Bartasevick, 1976) the invention relates to an upright grass trimmer for trimming relatively wide bands of grass. The trimmer employs a fixed multi-toothed bottom blade formed of the thin metal plate and four multi-toothed top blades, also formed of a thin metal plate, which act as two sets of top blades. The top blades are arranged for oscillatory motion and are phased to achieve near dynamic balance with reduced vibration, and achieve substantial load leveling. The top blades are driven by a battery-pack-powered D.C. motor through a clutch mechanism which absorbs abrupt blade stoppage and blade shock situations. A molded shroud is provided as a covering for the motor and other internal parts and also provides an elevated cavity having an open end for receiving the mating battery pack. The battery pack is removable for recharging and is held in the elevated cavity by a cantilevered strip member molded in the pack housing and which engages the cavity interior. Adjustable handle means are attached to the trimmer and electrical control means are mounted thereon. The operator remains in an upright position and pushes the trimmer along on wheels which are mounted rearward of the trimmer blades and on a skid which is mounted adjacent the trimmer blades but like DC (Direct Current) battery it easily get low in terms of battery capacity or storage of energy that suggested to put a source of power which is fixed to the equipment.

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In addition of related literature “A Multi-Purpose Grass Cutter” (Santos, 2006) was one of the bases of the present development of safety head trimmer that the innovation was a lawn mower type (Figure 2), that the device was equipped with a compromises an electric motor being mounted on a multi-wheeled carrier through a support bracket. A conventional bearing assembly is mounted on the forward portion of the said carrier a driven shaft is mounted on the forward portion of the said carrier a driven shaft is keyed to said bearing assembly. A driven pulley is keyed to the upper end of the said driven shaft.

Said motor having a drive pulley keyed to its driven shaft. The drive pulley of said being connected by an endless belt to said driven pulley.

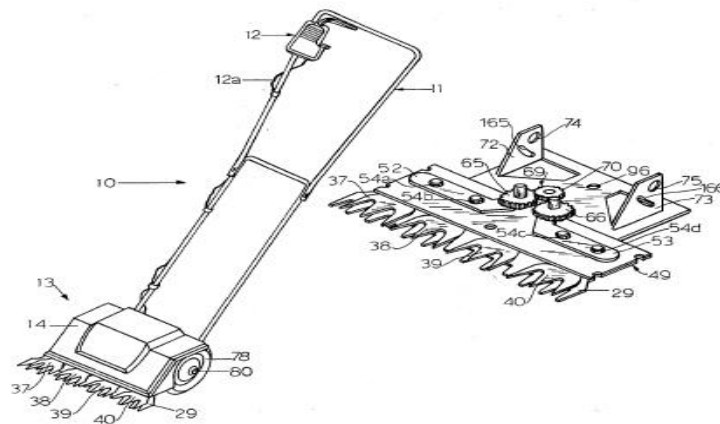


Figure 1. The Cordless Grass Trimmer Having Removable Battery Pack (Bartasevick, 1976)

A perforated plate is removable keyed to the lower end of the said driven shaft. A stand is bolted underneath the forward portion of the wheeled carrier aligned with the bearing assembly thereon which rotatable hold there on the lower end of said driven shaft through suitable bearings provided on said stand.

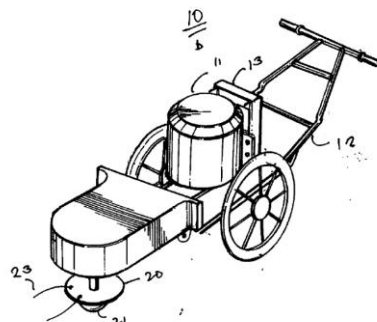
The plate may be replaced by a blade in the same manner as it is held on the drive shaft.

When the motor is turned on, it rotates the driven shaft through the pulleys, in turn, rotates the plate. The nylon cord in turn spin and is used as a grass cutter or lawnmower.

The nylon cord is heavy duty and there is no fear or worry about rocks, stumps, roots or hidden the hard object of any kind since there are no teeth to be broken as in conventional blades. For cutting bush wood, the plate is replaced by the blade.

It should be noted that the present grass cutter is electric driven so that is trouble-free and with fewer maintenance requirements compared to conventional type gas-portable grass cutter.

Furthermore, the components of the Multi-Purpose Grass Cutter (Santos, 2006) comprising a multi-wheeled carrier, a motor mounted on said multi-wheeled carrier, a shaft being driven by said motor and is mounted on the forward portion of said multi-wheeled carrier, a stand provided underneath the forward portion of said multi-wheeled carrier proximate the shaft, a perforated plate removable attach to said perforated plate, was also based on Gingerich' (1998), Electric Power Small Tractor (Figure 3).



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Figure 2. A Multi-Purpose Grass Cutter (Santos, 2006)

The Electric Power Small Tractor has independent electric motors for both driven wheels. The motors are combined with reduction gearboxes and the wheel is mounted on the output shaft of the gearbox. There is no mechanical axle connection. The power to the motor is controlled by an automatic controller as to wheel speed, power being increased/decreased in accordance with whether the wheels are under-running or overrunning relative to manually-controlled speed setting. Steering is controlled by a potentiometer on a mechanical steering wheel or by joystick control. The joystick may be on the tractor or remote. The drive system is suitable also for four-wheel drive.

The components of Electric Power Small Tractor having a conventional steering system. The non-driven wheels are mounted at the ends of an axle and are swivel-able about king-pins. A steering wheel turns the wheels for steering. The driven wheels are not mounted on a common axle. Each wheel has its own axle which is mounted directly in the respective gearboxes. The gearboxes are bolted to respective electric motors and to the chassis of the tractor. The (two-stage) gearboxes provide a gear ratio between the motor and the road wheel of 30:1. The gearboxes provide a firm mounting and a good bearing spread for the wheel axle bearings.

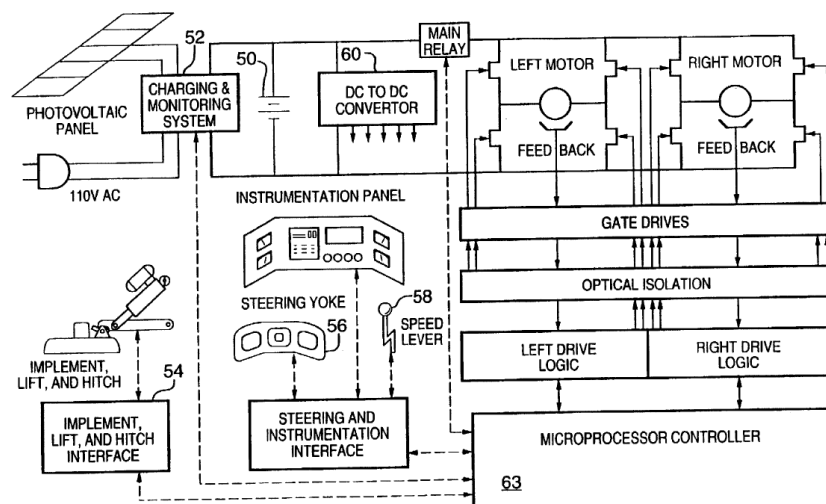


Figure 3. The Electric Power Small Tractor (Gingerich 1998)
Parameters as bases for Evaluation

Charging and discharging rate of the device's battery

A key parameter of a battery is the battery state of charge (BSOC). The BSOC is defined as the fraction of the total energy or battery capacity that has been used over the total available from the battery. Battery state of charge (BSOC or SOC) gives the ratio of the amount of energy presently stored in the battery to the nominal rated capacity. For example, for a battery at 80% SOC and with a 500 Ah capacity, the energy stored in the battery is 400 Ah. A common way to measure the BSOC is to measure the voltage of the battery and compare this to the voltage of a fully charged battery. However, as the battery voltage depends on temperature as well as the state of charge of the battery, this measurement provides only a rough idea of battery state of charge.

In many types of batteries, the full energy stored in the battery cannot be withdrawn (in other words, the battery cannot be fully discharged) without causing serious, and often irreparable damage to the battery. The Depth of Discharge (DOD) of a battery determines the fraction of the power that can be withdrawn from the battery. For example, if the DOD of a battery is given by the manufacturer as 25 percent, then only 25 percent of the battery capacity can be used by the load.

Nearly all batteries, particularly for renewable energy applications, are rated in terms of their capacity. However, the actual energy that can be extracted from the battery is often (particularly for lead-

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acid batteries) significantly less than the rated capacity. This occurs since, particularly for lead-acid batteries, extracting the full battery capacity from the battery dramatically reduced battery lifetime. The depth of discharge (DOD) is the fraction of battery capacity that can be used from the battery and will be specified by the manufacturer. For example, a battery 500 Ah with a DOD of 20 percent can only provide $500\text{Ah} \times .2 = 100\text{ Ah}$.

In addition to specifying the overall depth of discharge, a battery manufacturer will also typically specify a daily depth of discharge. The daily depth of discharge determined the maximum amount of energy that can be extracted from the battery in a 24 hour period. Typically in a larger scale PV system (such as that for a remote house), the battery bank is inherently sized such that the daily depth of discharge is not an additional constraint. However, in smaller systems that have relatively few days of storage, the daily depth of discharge may need to be calculated.

A common way of specifying battery capacity is to provide the battery capacity as a function of the time in which it takes to fully discharge the battery (note that in practice the battery often cannot be fully discharged). The notation to specify battery capacity in this way is written as C_x , where x is the time in hours that it takes to discharge the battery. In the above table, $C_{10} = xxx$ (also written as $C_{10} = xxx$) means that the battery capacity is xxx when the battery is discharged in 10 hours. When the discharging rate is halved (and the time it takes to discharge the battery is doubled to 20 hours), the battery capacity rises to xxx . The discharge rate when discharging the battery in 10 hours is found by dividing the capacity by the time. Therefore, $C/10$ is the charge rate. This may also be written as $0.1C$. Consequently, a specification of $C_{20}/10$ (also written as $0.1C_{20}$) is the charge rate obtained when the battery capacity (measured when the battery is discharged in 20 hours) is discharged in 10 hours. Such relatively complicated notations may result when higher or lower charging rates are used for short periods of time.

The charging rate, in Amps, is given in the amount of charge added the battery per unit time (i.e., Coulombs/sec, which is the unit of Amps). The charging/discharge rate may be specified directly by giving the current - for example, a battery may be charged/discharged at 10 A. However, it is more common to specify the charging/discharging rate by determining the amount of time it takes to fully discharge the battery. In this case, the discharge rate is given by the battery capacity (in Ah) divided by the number of hours it takes to charge/discharge the battery. For example, a battery capacity of 500 Ah that is theoretically discharged to its cut-off voltage in 20 hours will have a discharge rate of $500\text{ Ah}/20\text{ h} = 25\text{ A}$. Furthermore, if the battery is a 12V battery, then the power being delivered to the load is $25\text{ A} \times 12\text{ V} = 300\text{ W}$ (about the power is drawn by xxx). Note that the battery is only "theoretically" discharged to its maximum level as most practical batteries cannot be fully discharged without either damaging the battery or reducing its lifetime (Battery, 2017). Furthermore, the parameter in this part was utility model to the study of Yap (2016) entitled solar-driven desalination device. In the previous study, a battery was used as a storing device from the solar panel that drives the desalination device.

Acceptability of Device in terms of Design, Composition, and Operating Performance

The acceptability of the device in terms of design, composition, and operating performance was adopted from the study of Apolinario (2013) entitled "Electromagnetic saving device".

Design. The creation of a plan or convention for the construction of an object, system or measurable human interaction (as in architectural blueprints, engineering drawings, business processes, circuit diagrams, and sewing patterns). The design has different connotations in different fields. In some cases, the direct construction of an object is also considered to use design thinking. Designing often necessitates considering the aesthetic, functional, economic, and sociopolitical dimensions of both the design object and design process. It may involve considerable research, thought, modeling, interactive adjustment, and re-design (Brinkkemper, 1996).

Composition. The arrangement, process, and/or art of composing and constructing the device in a scientific manner (Apolinario, 2013). The device was evaluated according to (1) the quality of the materials being used conforms to the requirements of the equipment, (2) The equipment fabrication procedure is followed properly in assembling the equipment, (3) The structure's components are suitable for trimming the grass, (4) The fabrication of development of an auxiliary fan into solar power grass trimmer and the other components is a result of a skillful planning, (5) The unit can easily be installed and the components can easily be replaced or maintained.

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Operating Performance. The standard or prescribed indicators based on effectiveness, efficiency, and environmental responsibility (Baquilar, 2016). It was evaluated in terms of (1) The structure can trim the grass continuously for a certain period of time, (2) The system has a faster charging rate, (3) The system has a slower discharging rate based on the observed data, (4) The equipment is safe to use and less maintenance, (5) The system has a power surplus after operation.

Summary of the Reviewed Literature

The utilization of alternative energy particularly the solar energy motivated the researcher to venture into this developmental research. Several studies and utilization of solar energy and the design, composition, and operating performance of harnessing and increasing the performance of the previous inventions made the present study more feasible by making some innovation.

The review of the related literature was the basis for intellectualizing and making the layout plan as well as in assessing the development of equipment starting from the design, composition, and operating performance. The present equipment had three prior arts: The Cordless Grass Trimmer Having Removable Battery Pack of Bartasevick, (1976); A Multi-Purpose Grass Cutter by Santos (2006), and Electric Power Small Tractor Gingerich (1998), other related innovation and model of grass trimmer, also contributed in the conceptualization of the design, composition and its operating performance.

(11) Materials and Methods or Methodology

This study is developmental research .Developmental research is particularly important in the field of instructional technology, and has been defined as the systematic study of designing, developing, and evaluating instructional programs, processes, and products that must meet criteria of internal consistency and effectiveness (“Institution of Education Sciences”, 2012)

It further refers to the plan and structure of the investigation used to obtain evidence to answer research questions. The most common types of developmental research involve situations in which the product-development process is analyzed and described, and the final product is evaluated. (Calmorin, 2010).

The following criteria were considered in the development of the design of the solar grass trimmer: utilization of the renewable source of energy; stored energy; easy to operate; portable; and volt and amp indicator.

Utilization of renewable source of energy. The design used renewable energy in the form of solar energy. The energy generated from the solar panel as module then regulated and stabilized by solar charger controller PWM type is stored in two (2) units of 12 volts/7,000mAH battery.

Stored energy. The stored energy in two (2) units of 12 volts/7,000mAH battery can sustain the operating time of the solar grass trimmer up to three hours when it is fully charged.

Easy to operate. The solar grass trimmer is started through switch button thus making it easy to operate.

Portable. The device was made up of galvanized iron pipe joint together to construct the drive shaft of the grass trimmer, and an improvised aluminum frame which the solar panel was inserted for solar energy catchment that is worn as backpack.

Volt and Amp Indicator. The consuming rate of the battery is monitored with the use of volt and amp indicator attached to the device to operate the 24-volt auxiliary fan motor mounted in improvised steel drive shaft that was made to control in trimming the grass.

The design was actually the product of the continuous effort of the researcher in imagining and in visualizing the outlook of the design to come up with a device suitable to answer the comfortable way of manipulating the product.

The functions of the different parts of the system in the design plan preparation were stated consecutively.

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Solar Panel

Solar panel module having an economical frame for supporting solar panels in a high-density array and adapted for ready attachment to an improvised frame made of aluminum. The solar panel is made up of many solar cells linked together to form a circuit and is mounted in a frame. PV solar panels generate DC electricity, which must then be converted to AC electricity by an inverter (Peterpaul, 2016).

Solar Charger Controller

Charger controller limits the rate at which electric current is added to or drawn from electric batteries. It prevents overcharging and may protect against overvoltage, which can reduce battery performance or lifespan, and may pose a safety risk (Fowler, 2006).

Battery

Battery is a group of two or more cells connected together to furnish electric current (Fowler, 2006).

Digital Ammeter Voltmeter Dual Display

Digital volt and amp meter an electronic device mounted fixed in equipment that can monitor the volt and amp rates of the equipment by fixed installing to the aluminum case connected to the battery and 24 volt motor.

Switch

A small device that starts or stops the flow of electricity to something (such as a lamp or a machine) when it is pressed or moved up and down (Shefter, 2006).

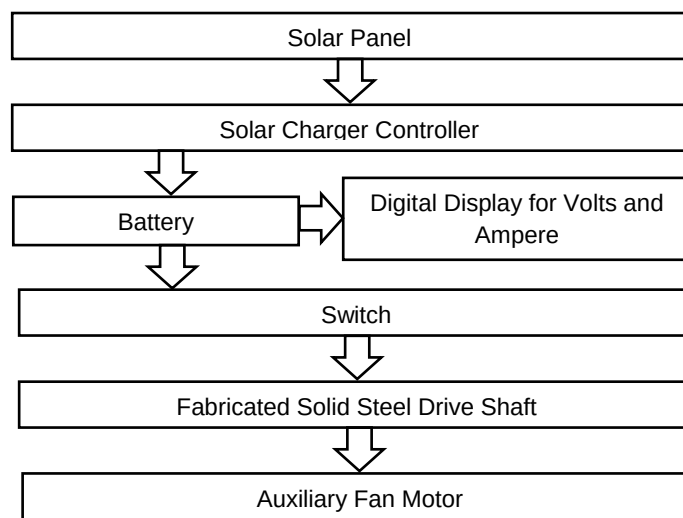
Fabricated Solid Steel Drive Shaft

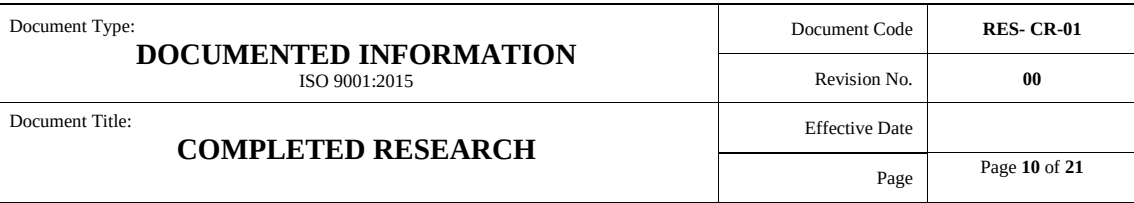
Made of galvanized iron pipe welded together were the operator handed the grass trimmer comfortable and where the auxiliary fan motor mounted at the bottom.

Auxiliary fan motor

A supplemental underground fan installed to increase the volume of air to a specified location for the purpose of controlling dust, methane, or air quality (Berman, 2006).

Figure 4 presents the block diagram of the components of the backpack solar powered grass cutter. With the use of a solar panel it helps to gain solar heat to be converted into energy and with the use of solar charger controller support to regulate and stabilized the charging into the charging rate capacity of the battery and with the use of the battery it help the solar charger controller to recognized the number of volts of the battery and at the same time operate the charger controller then directly connected the motor into the battery control by switch to turn it on and off.





The materials used in the research were: 24 Volts DC brushed motor, 12 volts 7,000mAH Battery, 10 watts Solar Panel 14inch. X 9.5inch diam., 7 meters stranded wires, 5ft. Galvanized Iron Pipe 1inch Ø, SPST switch, solar charger controller, digital display for volts and ampere, and 6ft. aluminum steel bar.

The tools and equipment used were soldering iron with lead and flux, multimeter, electric drill, long nose plier, screw driver, rotor razer, desoldering pump, diagonal cutting plier, welding machine, riveter, thermometer, tachometer and angle grinder.

The development of this innovation was a product of trial and error as to the selection of the materials to be utilized in making the different parts of the product. A design plan was made using a block diagram (Figure 4) prior to the construction. Hence, the following steps were sequentially prescribed: (1) A solar panel was mounted on improvised bag pack made of aluminum was prepared with the dimension of 38.5cm length, 50.5cm height, 13cm width. The solar panel was connected in series wiring that can sum up the voltage of the solar panel of 21.75 volts ($21.75\text{v}+21.75\text{v} = 43.5\text{volts}$). (2) A solar charger controller helps to regulate and stabilize the charging input by detecting the type of the battery and measure the charging capacity from the solar energy from the solar panel this helps the battery from being over charge was mounted inside the improvised aluminum case. (3) The two units of 12-volts 7AH (Ampere-Hour) battery was connected in series wiring to support the power voltage of 24-volts auxiliary fan motor was place inside the bag pack aluminum case. (4) Digital Ammeter Voltmeter dual display was placed beside the aluminum case of the solar panel to monitor the battery rate. (5) The switch was mounted near the handle grip to control the flow of the current to the auxiliary fan motor. (6) A pre-fabricated solid steel drive shaft was made of 1/2 inches; schedule 40 galvanized iron pipe welded together to serve as the main body of the grass trimmer where the 24-volts auxiliary fan attach at the bottom. (7) 24 volts/70 watts auxiliary fan motor was used to operate the equipment to trim the grass with an improvise head trimmer mounted with two cable ties.

Ghantt Chart

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The Development of Backpack Solar Power Grass Cutter

Figure 6 presents the pictorial diagram of backpack solar powered grass cutter from auxiliary fan including its different parts such as a solar panel, solar charger controller, digital volt/amp meter dual display, 2 units of 12volts battery, switch, steel drive shaft 24 volts auxiliary fan motor.

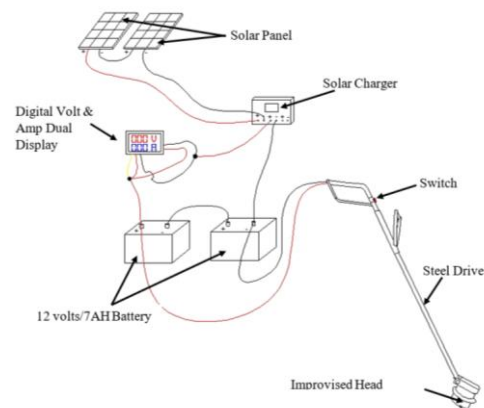


Figure 6. The pictorial diagram of the backpack solar power grass cutter.

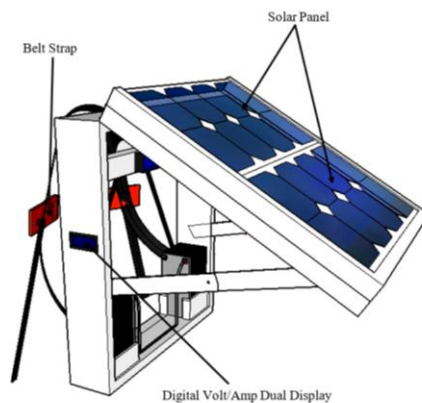
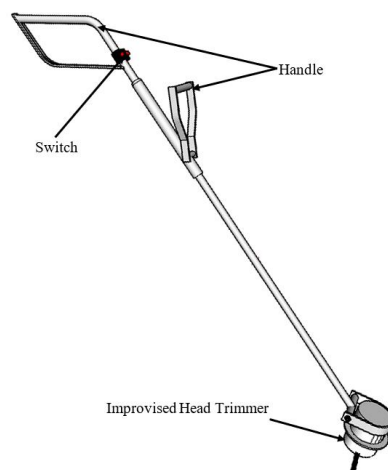


Figure 7. Isometric View of the Power Source of the Equipment as Backpack Design



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Figure 8. Improved Solid Steel Drive Shaft of Grass Cutter.

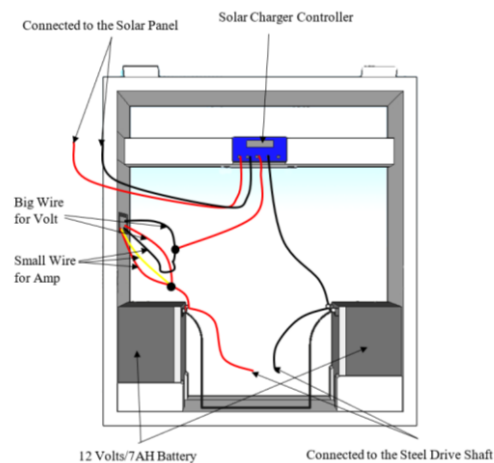


Figure 9. Wiring Diagram of the Equipment.

Evaluation Procedure

The development of solar powered grass cutter from the auxiliary fan was evaluated in terms of charging rate of the battery within 3 hours, 6 hours, and 12 hours in terms of voltage; discharging rate within 1 hour, 2 hours, and 3 hours considering the four by eight grass field area in terms of voltage and current and the level of acceptability of the development of solar powered grass cutter as to its design such as the design possesses innovation, the design is compatible with both gender to use, The design is suitable for cutting the grass, the parts and components are properly laid out and accurately assembled, the design provides precautionary measures for the safety of parts and its accessories. Composition such as the quality of the materials being used conforms to the requirements of the equipment, the equipment fabrication procedure is followed properly in assembling the equipment, the structure's components are suitable for cutting the grass, the fabrication of development of auxiliary fan as backpack solar power grass cutting and the other components is a result of skillful planning, the unit can easily be installed and the components can easily be replaced or maintained. Operating performance such as The structure can cut the grass continuously for a certain period, the system has a faster charging rate, The system has a slower discharging rate based on the observed data, the equipment is safe to used and less maintenance, The system has a power surplus after operation, was indicate using researcher's made observation sheet and evaluation sheet.

The testing process of the different electrical and components using evaluation and test instrument was done for a scientific result. At the college of Electrical Technology section and Electronics section for the availability of the testing instrument. Laboratory testing was done in the presence of electrical and electronics technology professors.

Series of testing was done to make the equipment useful. The trial and error method was applied and the theoretical ideas of some engineers and technologies were consolidated. Series of revisions were done in the exact proportion of solar powerer grass cutter attain the stable performance of the equipment. Additional electronic components were also installed for the refinement of the equipment performance. The pilot testing and revision were successful for the duration of two months.

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The system was subjected to evaluation of its suitability in terms of its design, composition and operating performance. The researcher conducted the evaluation at Capiz State University-Main Campus, Fuentes Drive Roxas City, Capiz. The evaluators were convenience selected because they are experts on the technology, have considerable background on the technology and practicing professionals in related studies.

The evaluators were composed of 10 industrial technology academe from electrical and electronics technology, 3 registered electrical engineers, 7 casual workers from Department of Public Works and Highways DPWH, 3 local farmers and 7 utility aide.

The following were the parameters considered.

1. Testing for Current

Current testing was done using volt and amp dual display to determine the amount of current that flows in the system with an electrical load. The low current measurement of the output contributed to low energy consumption. The electrical professor together with evaluators, experts, and the researcher monitored the actual current drawn by the system as the basis of scientific evaluation on current drawn.

2. Testing for Voltage

The voltage testing was done using a volt and amp dual display to determine the voltage input of the system as the primary source.

Parameters to be Analyze

Charging Rate:

Here is the formula to calculate for the operation of the develop solar powered backpack charger.

$$\text{Battery Charging Time (in Hours)} = \frac{\text{Battery Capacity (in mAH)}}{\text{Charging Current (in mA)}}$$

Required Charging Current for battery

$$\text{mA} = \text{Battery mAh} \times 10\%$$

Load consumption and Discharging

$$\text{mAh} = \frac{\text{Watts}}{\text{Battery Voltage}}$$

Discharging Rate with and without solar panel

$$\text{Discharge Rate} = \frac{\text{Total Battery Capacity (in mAH)}}{\text{Load Consumption (in mAH)}}$$

Instrumentation

A standardized evaluation sheet was made using a five-point Likert scale was utilized. Every item in the questionnaire was provided with 5 choices:

Scale of Means

Score	Response Category	Score Interval
5	Strongly Agree	4.21 – 5.00
4	Agree	3.41 – 4.20
3	Uncertain	2.61 – 3.40
2	Disagree	1.81 – 2.60
1	Strongly Disagree	1.00 – 1.80

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However, the scores were the basis for the level of acceptability through scaling of means with verbal interpretation and description.

Data Gathered

The data gathering was done through the series of actual observation and manipulation in the development of solar powerer grass cutter and determining of the level of acceptability of the backpack solar powerer grass cutter as to its design, composition and operating performance as evaluated by the group of experts.

An actual observation was conducted to find out the acceptability of backpack solar powerer grass cutter. It was observed and the data gathered were recorded in the observation sheet.

This study was evaluated by thirty (30) evaluators, ten (10) from academe, and seven (7) from utility aide, seven (7) experts from industry DPWH casual worker three (3) from local farmers, three (3) from engineering industry. They were selected using a combination of purposive sampling and convenience sampling. Convenience sampling (sometimes known as grab or opportunity sampling) is a type of non-probability sampling which involves the sample being drawn from that part of the population that is selected because it is readily available and convenient.

While in purposive sampling, the researcher chooses the sample based on who he thinks would be appropriate for the study. This was used primarily when there is a limited number of people that have expertise in the area being researched (Castillo, 2009).

The evaluators were asked to check each item in the questionnaire for the level of acceptability as to its design, composition and operating performance.

Cost Analysis

Cost analysis of materials used in the backpack solar powerer grass cutter.

Table 3. Cost of materials used in the construction of development of solar power grass cutter.

No Qty.	Unit	Description	Unit value	Total Value
1	pc.	24 volts auxiliary fan motor	600.00	600.00
2	pcs.	12 volts 7AH battery	350.00	700.00
2	Pcs.	10 watts Solar Panel 14'x9.5"	450.00	900.00
4	mts.	Stranded wire	22.00	88.00
1	Pc.	5' Galvanized Iron Pipe 1"Ø	150.00	150.00
3	ft.	3' pla plates	20.00	60.00
1	pc.	Belt Strap	360.00	360.00
6	ft.	Aluminum Steel square bar	20.00	120.00
1	Pc.	Switch	25.00	25.00
1	Pc.	Solar Charger Controller	350.00	350.00
1	Pc.	Digital Volt/Amp Meter Dual Display	130.00	130.00
Total				3483.00

Commercial Accessories:

Cost of Materials	3483.00
Labor	+ 1393.20
Total for the equipment only	4876.20

(12) Results and Discussions

Charging Rate of the Backpack Solar Power

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Grass Cutter in terms of Milliamp Hour (mAH)

Table 4 shows the charging rate of the developed backpack solar powerer grass cutter in terms of milliamp hour. The standard battery capacity charging rate is 10% of the total battery capacity and found out that it has a 700mA charging rate and it takes 10 hours to fully charge the battery from solar panel. However, the experiment only done during sunny days and sun hits its maximum heat capacity during 9:AM to 4:00PM.

Table 4. Charging rate of developed backpack solar powered grass cutter

Battery Capacity	Charging Current	Charging Capacity	Charging Power
7,000mAH	700mA	10%	16.8 Watts
Charging rate Temperature		700mAH	29°C-30°C

Discharging Rate while in Operation without Solar Panel in terms of Voltage

Table 5 shows the discharging rate of the developed backpack solar powered grass cutter without solar panel in cutting grass field of 50 sq. meters

The storage battery had a 24.7 initial volts with 7,000mAH, the load consumption was 37.68 watts and it was divided by the initial voltage resulting to have a 1,525mAH. Since we have a load consumption in milliamp hour it was then divided to the total capacity of the battery which is 7,000mAH resulting to 4.59 hours of continuous operation before the battery discharge.

Therefore using low AH (Ampere-Hour) battery only without solar panel mounting is not sufficient to trim the grass for a long time using auxiliary fan motor as an improvised head cutter.

This result was also in consonance with other batteries which are seldom fully discharged and manufacturers often use the 80 percent depth-of-discharge (DoD) formula to rate a battery, meaning that only 80 percent of the available energy is delivered and 20 percent remains in reserve or about 7 volts is still remaining in each battery. Cycling a battery at less than full discharge increases service life and manufacturers argue that this is closer to a field representation because batteries are commonly recharged with some spare capacity left ("Basics About Discharging", 2015).

Table 5. Discharging Rate while in operation without solar panel in terms of mAH.

Initial Voltage	Discharging Rate		
	Battery Capacity (mAH)	Load Consumption	Discharge Rate
24.7 volts	7,000mAH	37.68 Watts	4.59 hours
Discharging rate/hour	1,525mAH		

Discharging Rate while in Operation with Solar Panel Connected

Table 6 shows the discharging rate of the backpack solar powered grass cutter without solar panel support within 50 sq meters initial area to measure the capacity wattage of load consumption with the solar panel connected.

Since the solar panel is connected the working device the researcher theorized that the extra power to operate the grass cutter came from solar panel and battery. The result of the measured parameters

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shows that the grass cutter consume 13.853 watts only in a rate of 650mAH of battery. Since the battery capacity is 7,000 mAH it takes about 12.5 hours of continuous operation before the battery runs out.

Table 6. Discharging rate while in operation with solar panel connected

Initial Voltage	Discharging Rate		
	Battery Capacity (mAH)	Load Consumption	Discharge Rate
24.7 volts	7,000mAH	13.835 Watts	12.5 hours
Discharging rate/hour		560 mAH	

Acceptability of the Development of Backpack Solar Powered Grass Cutter in terms of Design, Composition, and Operating Performance

Table 9 shows that the overall mean of the acceptability of development of backpack solar powerer grass cutter in terms of its design, composition and operating performance was 4.74. This overall mean which was the average of the three grand means implied that the evaluators had evaluated the equipment as "Very Acceptable".

Further examination of the data shown that in all three constituents, the mean ranged from 4.21 to 5.00, all of which had a verbal interpretation as "Very Acceptable". The highest mean of 4.80 in terms of design followed with 4.79 in terms of composition and lastly 4.62 in terms of operating procedure.

The result involves that the design includes the provision and pre-occupational measures for the safety of detail parts and its fittings that suits for cutting the grass that you can move according to your desire due of the source of power to operate the equipment is design for easy to carry through bag pack. In the same manner how to utilize solar power to generate electricity to operate the equipment.

Furthermore, the composition of the equipment was carefully and skillfully planned with the available materials used to construct the equipment, as a result, it implies that the equipment can operate using free energy and renewable without any harm both to the environment and human where it conforms to the suggestion of Bartasevick, 1976 entitled "The Cordless Grass Trimmer Having Removable Battery Pack" that having a fixed charging power supply as alternator to the battery while discharging is much appreciated.

In the same manner, the result implies that the equipment can operate continuously within 3-4 hours, however, that the charging rate needs for some improvement because it takes about 2 days if the heat of the sun is the primary source to charge the storage device to become a full charge. The result further proves that the solar powerer grass cutter from auxiliary fan suit the operating performance since that even pro-grass trimmer from the DPWH as one of the evaluator said that even two strokes engine don't long last for a certain time because of the engines becomes hotter when it reaches the overheat point that they need to shut down the engine and wait about half hour to cool down the engine and start doing their job again.

Moreover, the development of solar power grass cutter was "Very Acceptable" in terms of its design, composition and operating performance as supported by the reviewed literature of this study.

Table 9. Acceptability of the backpack solar powerer grass cutter in terms of design, composition, and operating performance.

Variables	Mean	Verbal Interpretation
Design	4.80	Very Acceptable
Composition	4.79	Very Acceptable
Operating Performance	4.62	Very Acceptable
Overall Mean	4.74	Very Acceptable

(13) Conclusions and Recommendations

Summary of Findings

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Solar energy becomes much more popular nowadays. Even electrical companies are using solar power for their power source. As a researcher, he looked forward to how he can manipulate solar energy as an alternative source of backpack solar powered grass cutter. The main objective was to determine the acceptability of the developed backpack of grass cutter utilizing the solar power as an alternative energy.

Specifically, this study aimed to: The study aimed to: (1) determine the charging rate of the Developed Backpack Solar Powered Grass Cutter in terms of milliamp hours (mAH); (2) determine the capacity of the device in discharging rate in terms of wattage used and the milliamp hour (mAH) of battery of the solar power grass cutter without solar and with solar charging; (3) determine the level of acceptability of the Developed Solar Power Grass cutter as to its design, composition, and operating performance.

Developmental method of research was used in the design of grass cutter utilizing solar power as an alternative source of energy to operate the equipment. The construction of the equipment was a product of trial and error. The data were gathered through observation and through the use of evaluation sheet during actual manipulating of the equipment by the local farmers, pro-users such as Department of Public Works and Highways (DPWH) casual workers. The evaluators were experts in the field of gardening, electronics, mechanical and allied fields.

The findings of the study was:

The charging rate of the developed backpack solar powerer grass cutter in terms of milliamp hour. The standard battery capacity charging rate is 10% of the total battery capacity and found out that it has a 700mA charging rate and it takes 10 hours to fully charge the battery from solar panel. However, the experiment only done during sunny days and sun hits its maximum heat capacity during 9:AM to 4:00PM.

The discharging of battery when tested in the area of 50 sq. meters when it was not connected to the solar panel was 1,525mAH since the the load consumption was 37.68 watts and it was divided to the voltage of the battery. Moreover, the load consumption (in milliamp hour) was divided to the total capacity of the battery which is 7,000mAH resulting to 4.59 hours of continuous operation before the battery discharge.

The discharging of battery when tested in the area of 50 sq. meters when it was connected to the solar panel was 650mAH with a load consumption of 13.853 watts only. The researcher theorized that the solar panel helps the excess wattage to run the grass cutter. Since the battery capacity is 7,000 mAH it takes about 12.5 hours of continuous operation before the battery runs out.

Furthermore, the backpack solar powerer grass cutter was "Very Acceptable" in terms of its design, composition, and operating performance.

Conclusions

On the basis of the findings of the study, the following conclusions were drawn:

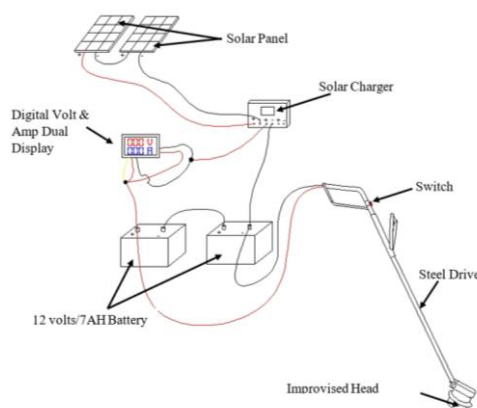
1. The charging rate of the backpack solar powerer grass cutter was 700mAH and it takes 10 hours to fully charge the battery when not used.
2. The discharging rate of the developed backpack solar powered grass cutter without solar panel was 1,525mAH and it takes 4.59 hours of continuous used to drain the battery, however when the device was connected to solar panel the discharge rate of the battery was only 560mAH and it takes 12.5 hours to fully drain the battery. The researcher theorized that the excess power has been compensate by the solar panel to run the motor which means that when the solar panel is connected more than 60% of of the power comes from the solar panel resulting to the prolong used of solar grass cutter.
3. The backpack solar powerer grass cutter has a very acceptable design, very acceptable composition, and very acceptable operating performance. Therefore, the innovation made in the design, the use of materials and integration of the different parts and its capacity to its purposive essential have gained approval from the evaluators and they considered it with greater attributes. Thus, it is safe and very convenient in terms of eco-friendly and humanitarian attributes.

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Recommendations

1. For further improvement of the study, the recommendations below are considered by the researcher. Some of the recommendations are the insights imparted by the evaluators.
2. The charging system may be modified to increase the charging rate of the equipment by using MPPT or maximum power point tracking that included in charging controller use for extracting maximum available power from the solar panel under certain sunlight is recommended to have a stable supply of electricity during a charging rate.
3. The storage battery of the equipment can upgrade into higher mAH or milliamp hour so that it can long last up to 6 to 7 hours.
4. The use of light-weight materials such as alloy and stainless steel is recommended so that the equipment can be used by both genders.
5. Improve the winding of copper of the DC motor to add the rotation per minute that can make faster the rotate of the improvised head cutter that can trim more grass in a single hit.
6. Electrical/electronic students may be encouraged to perform different electronic jobs, innovations, repair, and troubleshooting using the equipment for them to be more productive.
7. Homeowners/Local Farmers are also encouraged to use the equipment to lessen their maintenance consumption to the traditional grass cutter.
8. The resort, golf course, amusement farm owners are encouraged to use the equipment to utilize free energy available in the environment. By this, it will minimize the pollution caused by the use of the traditional fuel power grass cutter.
9. According to one of the Engineer specialist engineer in electronic communication engineering or ECE recommend to develop a program that the power source has detector such as a buzzer or Light Emitting Diode LED to notify by the operator that gives advance signal whichever the power source is a full charge or low since the power source is behind of occurrences.
10. Other researchers are encouraged to conduct further studies related to the utilization of solar energy for innovating agricultural equipment as for improvement and development of new inventions.
11. Furthermore, the recommendation suggested can be used to develop the study in the near future.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.) The product is a Backpack Solar Powered Grass Cutter with 2 units solar panel, solar charge controller, 2 units 12 volts battery, digital display and DC operated grass cutter.



This paper presents the fabrication and working of a developed backpack solar grass cutter. In this work, we have developed a solar-powered grass cutter and thus saved energy by decreasing air pollution and reducing labor cost. In the this model, nylon (polyamide) was used. Due to its high environmental impact, it was the least expensive cutter used by the motor. We have utilized a solar panel, solar charger controller to control and protect battery from overcharging. The solar panel is attached to a backpack style to reduce burden to user and give a shade to the user from direct sunlight. DC motors (7000–13,000 RPM, starting current 1.7 A) are used to move the solar grass cutter, and one DC blade motor is used to cut the grass quickly. With current technology, this new product is designed as lightweight and easily to control like common grass cutter. The switch to turn on the motor is located to handle was

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designed to be quick release when the user did not touch the switch the motor will not rotate, this will ensure the safety of user and other people from accident. The battery is 12 volts 7AH connected in series to produce 24 volts to make the motor rotate much faster and more torque than on the 12 volts input. The handle is made of steel pipe which the wire runs inside it and also design like telescopic. This means that the user can be adjusted according to the user's height.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

Sample Economic Analysis

REVENUE

Sales

DIRECT COST

Direct Charges- Wages & Labor

Direct Charges - Materials & Supplies

Direct Charges – Depreciation

Direct Charges - Repair/Maintenance & Fuel/Oil

COST OF TECHNOLOGY PRODUCED

GROSS PROFIT

4,856.00

PRODUCTION EXPENSES

Salaries & Wages

Lights and Water

Telecommunications

Transportation

Office Supplies/Supplies

Permits & Licenses

Rentals and Expenses

Miscellaneous

Advertising & Promotion

Freight Expense

Total Production Cost

NET INCOME

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$$\text{ROI (\%)} = \frac{\text{Net Income}}{\text{Total Project Cost}}$$

$$\text{Payback Period (Years)} = \frac{\text{Total Project Cost}}{\text{Average Annual Profit}}$$

(17) Acknowledgments

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Submitted by:

REMAR D. APOLINARIO, EdD

Signature of Faculty Concern

AMEL L. MAGALLANES, PhD

Signature of Faculty Concern

Recommending Approval:

ENGR. JOEL C. VILLARUZ, PhD

Dean, CEAT

Approved:

MA. CONSUELO Y. YAP, EdD

Research Chair

Date: _____